

Vibe Coding & Prompt Engineering

Reimagining a Library Management System using AI

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Passionate about AI, Software Engineering, and building rapid prototypes.

Software Intern

Vodafone



Software engineering internship focusing on telecommunications systems.

Software Intern

P&G / Always



Developing enterprise software solutions and optimizing workflows.

AI Tutor

xAI / X Corp



Training state-of-the-art AI models and refining generated outputs.

Model Trainer

Grok Ecosystem



Specializing in prompt engineering and conversational logic for Grok.

The Challenge: Modernizing the Stack

Legacy System

- ✗ PHP Pages & MySQL Database
- ✗ Heavy XAMPP dependency
- ✗ Manual database import processes
- ✗ Complex local configuration

The AI Solution

- ✓ React (Vite) Frontend
- ✓ Node.js (Express) API Backend
- ✓ Zero-config local JSON storage
- ✓ Generated entirely by CODEX prompts

What is Vibe Coding?

Software development is fundamentally changing. We are moving away from writing line-by-line syntax to directing the architectural flow.

You provide the "**vibe**"—the core logic, the user story, and the visual structure. The AI handles the boilerplate, the routing, and the heavy lifting.

You are no longer just a coder; you are the Director.

Principles of Prompt Engineering



1. Be Specific

Define the exact stack and boundaries. Tell the AI what not to use (e.g., "does not require MySQL or XAMPP").



2. Iterate

Don't build everything at once. Establish the base architecture first, verify it works, then layer complex features on top.



3. Provide Context

Explain the "why". Telling the AI the user story helps it understand edge cases and generate better logic automatically.

The AI Architect: CODEX



Full Stack Translation: We used CODEX to completely translate the monolithic PHP codebase into decoupled React components and Express routes.



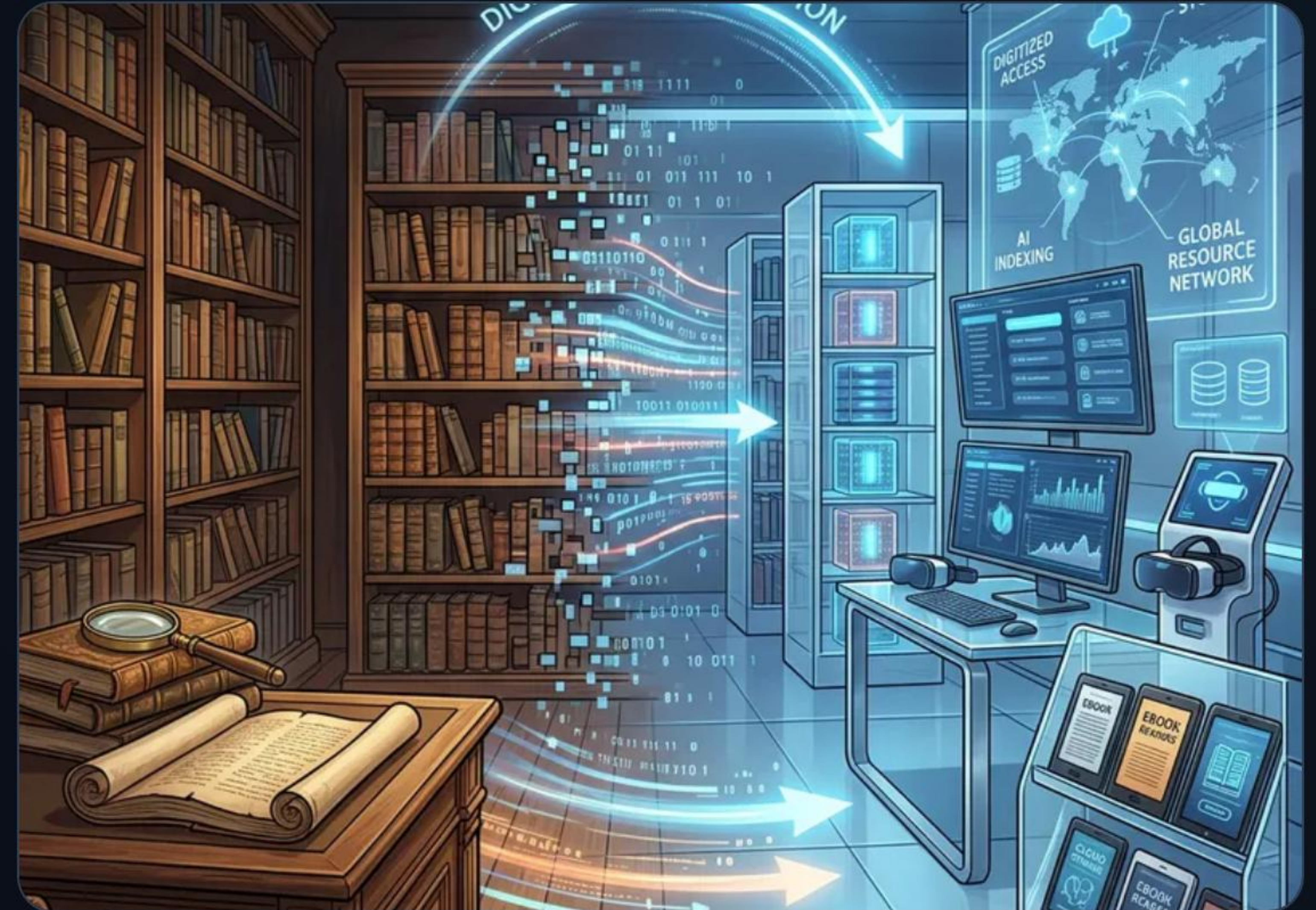
Secure Implementation: JWT authentication and file upload logic (Multer) were implemented flawlessly through pure natural language instruction.



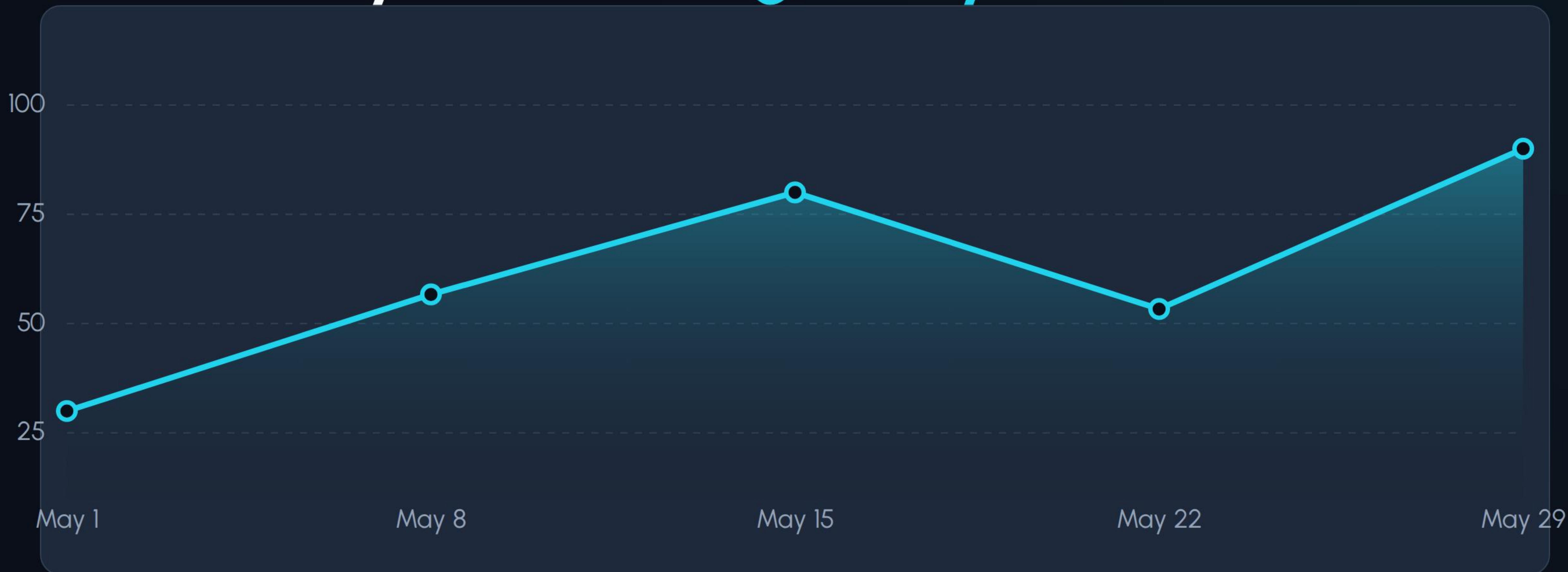
Let's trace the History: We will walk through the actual text file showing the real conversation and prompts that built this software from scratch.

Generated Architecture

-  **Frontend (React + Vite):** Responsive design, modern components, JWT local storage auth.
-  **Backend (Node + Express):** REST API handling user sessions, book management, and file streams.
-  **Storage (JSON + FS):** `library.json` acts as the persistent datastore, while physical PDFs and images are stored securely on the local disk.
-  **Role-Based Access:** Strict segregation between Admin Dashboard features and Student workflows.



Case Study: Generating Analytics



Prompt: "Add a history record for purchased books... and put the total numbers in the dashboard."
The AI autonomously generated the event tracking models, the API filtering logic, and the UI rendering.

Case Study: UI & Logic Refinement

"In the Books tab for the students, there is a problem: the frontend is not good... Put only 3 books per row and fix the component sizes and look."

— Iterative bug fixing without writing CSS



Instead of inspecting elements and tweaking CSS flexbox manually, we described the desired visual state.



The AI automatically implemented responsive design limits (Desktop: 3, Tablet: 2, Mobile: 1).

Let's see the project **together!**

Live Walkthrough & Real-Time AI Updates

Questions?

Thank you for your time.



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Image Sources



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Image Sources



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